Signals & Systems

Workbook

Electrical Engineering
Electrical & Electronics Engineering

GATE & ESE

unacademy

Signals & Systems

Workbook

Electrical Engineering

Electrical & Electronics Engineering

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Edition : S&S-2024/2025

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GATE Syllabus

Electrical Engineering (EE) : Representation of continuous and discrete-time signals, Shifting and scaling operations, Linear Time Invariant and Causal systems, Fourier series representation of continuous periodic signals, Sampling theorem, Applications of Fourier Transform, Laplace Transform and z-Transform.

ESE Syllabus

Electrical Engineering (EE) : Representation of continuous and discrete-time signals, shifting and scaling operations, linear, time-invariant and causal systems, Fourier series representation of continuous periodic signals, sampling theorem, Fourier and Laplace transforms, Z transforms, Discrete Fourier transform, FFT, linear convolution, discrete cosine transform, FIR filter, IIR filter, bilinear transformation.

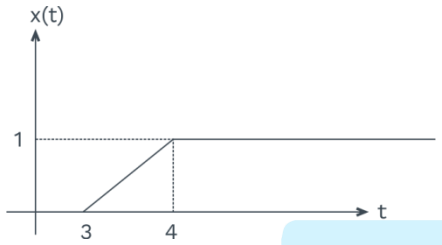
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1 Classification and Signals

Practice Questions

Q.1 The signal x(t) shown in figure expressed in terms of unit step and unit ramp function.



(A) r(t – 3) + r(t – 4)

(B) r(t – 3) – r(t – 4)

(C) r(t – 3) + u(t – 4)

(D) r(t – 3) u(t – 3) + r(t – 4) u (t – 4)

Q.2 Consider the integral

$$y(t)=\frac{1}{6}\int_{-\infty}^{\infty} x(\tau)[\delta(\tau-3)+\delta(\tau+3)]d\tau, y(t)$$

can be simplified to:

(A) $\frac{1}{2}[x(3)+x(-3)]$ (B) $\frac{1}{2}[x(3)-x(-3)]$

(C) $\frac{1}{6}[x(-3)+x(3)]$ (D) $\frac{1}{6}[x(3)-x(-3)]$

Q.3 Consider the sequence

$$x[n]=[-2+2j, \underset{\uparrow}{2j}, -4]$$

The conjugate anti-symmetric part of the sequence is:

(A) [1 + j, 2j, – 1 + j]

(B) [1 + j, 2j, – 1 – j]

(C) [3 + j, 0, – 3 – j]

(D) [1 + j, 2j, 2j]

Q.4 Find the even and odd components of unit step signal in terms of $u_e(t)$, $u_0(t)$ is. MSQ

(A) $\frac{u(t)+u(-t)}{2}, \frac{u(t)-u(-t)}{2}$

(B) $\frac{1}{2}, \frac{1}{2}\text{sgn}(t)$

(C) 1, sgn(t)

(D) u(t) + u(–t), u(t) – u(t)

Q.5 For a periodic signal

$$x(t)=30\sin 100t+10\cos 600t+6\sin\left(400t+\frac{\pi}{2}\right)$$

the fundamental frequency is rad/sec is:

(A) 200 (B) 600

(C) 100 (D) 400

Q.6 A signal x(t) = 2 cos(20πt) + 3 cos(60πt) is sampled at 120 Hz. Then the fundamental period of x[n] is:

(A) 4 (B) 16

(C) 8 (D) 12

Q.7 Given $x[n]=\begin{cases}2; & n=0\\ 1; & n=\pm1\\ 3; & n=\pm2\\ 4; & n=\pm3\\ 11; & n=\pm4\\ 0; & \text{otherwise}\end{cases}$ MSQ

Such the following options for x[3n]

(A) x[3n] = {4, 2, 4}

(B) 25 non-zero sample

(C) x[3n] = {11, 0, 0, 4, 0, 0, 3, 0, 0, 1, 0, 0, 2, 0, 0, 1, 0, 0, 3, 0, 0, 4, 0, 0, 11}

(D) 3 non-zero sample

Q.8 The value of integral $\int_{-5}^{+10} e^{-2t}\,\delta(2t-1)dt$ is equal to _____.

Q.9 The sum of two or more sinusoidal signal is -

(A) always periodic

(B) always periodic for discrete signal

(C) non-periodic

(D) periodic if it satisfied certain condition

Q10 Suppose $x_1[n] = \beta\, 0.25^n\, u[n]$, where β is a positive real number and $x_2[n] = 1.5$ only for $n = 0$. What is the value of β if energy of $x_1[n]$ and $x_2[n]$ is equal?

(A) 1.06 (B) 1.18

(C) 1.5 (D) 0.93

Q.11 Consider a complex exponential sequence $e^{j\omega.n}$ with frequency ω_0. Suppose $\omega_0 = 2$, then

(A) such a sequence is periodic

(B) such a sequence is periodic for certain value of 'N'

(C) such a sequence is never periodic

(D) such a sequence is semi periodic

Q.12 Two periodic signals x(t) and y(t) have the same fundamental period of 3 seconds. Let z(t) = x(– t + 2) + y(2t + 3). The fundamental period of z(t) in seconds is :

(A) 1 (B) 3

(C) 1.5 (D) 4

Q.13 Compute the value of $t\,\square(t)$

(A) $\frac{1}{2}$ (B) 1

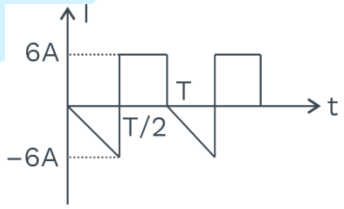
(C) 0 (D) ∞

Q.14 A current impulse of $5\delta(t)$ is forced through a capacitor C. The voltage $V_c(t)$ across the capacitor is given by:

(A) 5 C u(t) (B) $\frac{5}{C}u(t)$

(C) $-\frac{5}{C}u(t)$ (D) $-5Cu(t)$

Q.15 The rms value of the periodic waveform given in the figure is:



(A) $2\sqrt{6}A$ (B) $6\sqrt{2}$ A

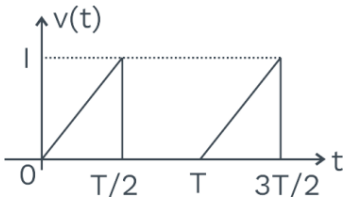
(C) $\sqrt{\frac{7}{2}}A$ (D) $\sqrt{\frac{4}{3}}A$

Q.16 The RMS value of the resultant current in a wire which carries a dc current of 10A and a sinusoidal alternating current of peak value 20A is:

(A) 1.732 A (B) 17.32 A

(C) 14.1 A (D) 10 A

Q.17 For the triangular wave shown in the figure, the RMS value of the voltage is equal to:



(A) $\sqrt{\frac{1}{3}}V$ (B) $\sqrt{\frac{1}{24}}V$

(C) $\sqrt{\frac{2}{3}}V$ (D) $\sqrt{\frac{1}{6}}V$

Q.18 A zero mean random signal is uniformly distributed between –a and +a and its mean square value is equal to its variance. Then, the rms value of the signal is-

(A) $\frac{a}{\sqrt{3}}$ (B) $a\sqrt{3}$

(C) $a\sqrt{2}$ (D) $\frac{a}{\sqrt{2}}$

Q.19 The energy of (t–1) u(t–1) – (t–2)u(t–2) – (t–3)u(t–3) + (t–4)u(t–4) is _____ .(up to two decimal points)

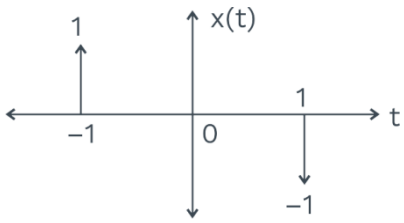
(Range from 1.65 to 1.67)

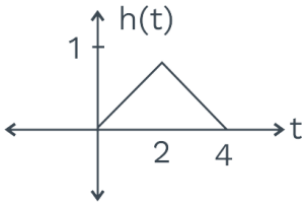
Q.20 The power of the signal $x(t) = 4\cos\left(30\pi t - \frac{\pi}{2}\right) + 6\cos(10\pi t)$ is

(A) 5 (B) 52

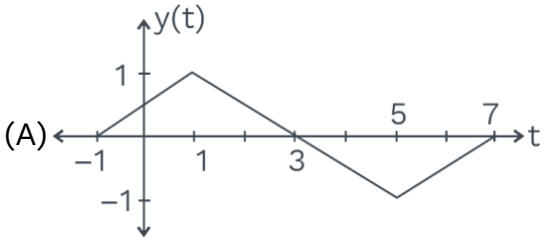
(C) 20 (D) 26

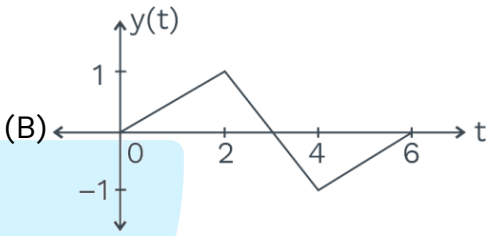
Q.21 The input signal x(t) is convoluted with h(t) to yield y(t)

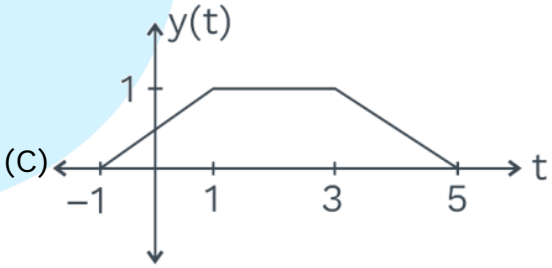


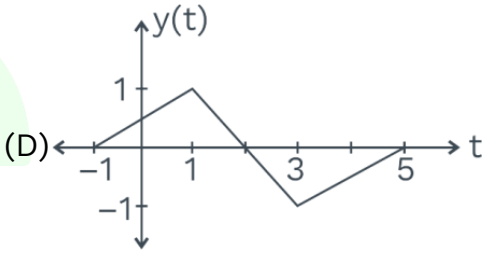


Which of the following figure represent the output y(t)?

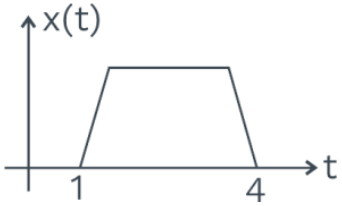
(A)


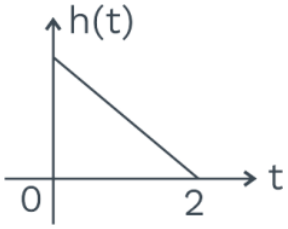
(B)


(C)


(D)


Q.22 Let x(t) be input a linear time invariant system and h(t) is the impulse response of the system.





The output of the system exists in the time interval.

(A) $1 < t < 6$ (B) $0 < t < 5$

(C) $0 < t < 4$ (D) $1 < t < 5$

Q.23 Given that $y(t) = x(t) * h(t)$ and $g(t) = x(5t) * h(5t)$. The relation between $y(t)$ and $g(t)$ is given by:

$g(t) = By(At)$

Find A and B

(A) 5, 5 (B) $\frac{1}{5}, 5$

(C) $5, \frac{1}{5}$ (D) $\frac{1}{5}, \frac{1}{5}$

Q.24 A discrete time signal is given as

$$x(n) = \cos\left(\frac{n\pi}{3}\right)(u(n) - u(n-6))$$,.

Determine the energy of the signal.

Q.25 Determine the fundamental period of following given discreate signal,

$$x(n) = \sin\left(\frac{6\pi}{7}n + 1\right)$$

2 Classification and Systems

Q.1 The system is defined by y(t) = 2x(t) + B. The system is -

(A) linear (B) linear if B > 0

(C) non-linear (D) linear if B < 0

Q.2 Let the system be defined by y(t) = x(t) cos $\omega_0 t$. The system is -

(A) linear if x(t) is periodic

(B) non-linear

(C) linear if x(t) exist for t > 0

(D) linear

Q.3 Let the system be y(t) = x(t) x(t–1). The system is -

(A) linear

(B) linear if both the signals are causal

(C) linear if both the signals are anti-causal

(D) non-linear

Q.4 The system is characterized by the equation y(t) = sin {x(t)}. The system is -

(A) non-linear

(B) linear

(C) linear if the signal becomes causal

(D) non-linear if signal becomes causal

Q.5 The system is characterized by y(t) = x(log t). The system is -

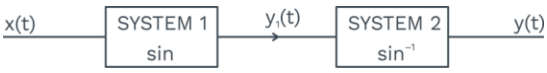
(A) non-linear

(B) non-linear for x{log t} is negative

(C) linear

(D) linear for x {log t} is negative

Q.6 Let there be two systems be cascaded. The two system which are cascaded are non-linear, then the overall response of system is:

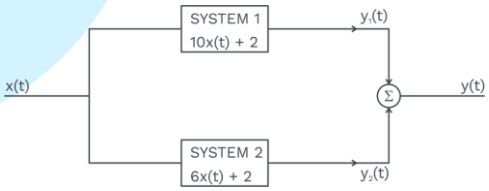


MSQ

(A) linear (B) non-linear

(C) $y_1(t)$ is linear (D) $y_1(t)$ is non-linear

Q.7 When two non-linear blocks are cascaded in parallel, the overall response is:



(A) linear (B) non-linear

(C) Non causal (D) Dynamic

Q.8 The system is characterized by $y(t) = \frac{d}{dt}x(t)$ then the system is

MSQ

(A) Non-linear (B) Time variant

(C) Time invariant (D) Linear

Q.9 The system is defined by $y(t) = \int_{-\infty}^{2t} x(\tau)\,d\tau$ the system can be defined as.

(A) Time invariant

(B) Time variant

(C) Time invariant if t > 0

(D) Time variant if t > 0

Q.10 Let the input signal be x[n] and the output of the system is defined by y[n] where y[n] = x[n] g[n]. The system will be-

(A) Time variant (B) Dynamic

(C) Time invariant (D) Non-causal

Q.11 Consider a system with input x[n] and output y[n] is related as y[n] = x[n]{g[n] + g[n – 1]} where; g[n] = 1 for all n. State if system is. MSQ

(A) Linear

(B) Time Invariant

(C) Time variant

(D) Non-causal

Q.12 A system with an input x(t) and output y(t) is described by the relation $y(t) = t^2x(t)$ the system is

(A) Linear and time invariant

(B) Linear and time varying

(C) Non-linear and time varying

(D) Non-linear and time invariant

Q.13 An audio amplifier is a system that takes the input x(t) and produces output y(t), shows the properties of system- MSQ

(A) Linearity

(B) Instability

(C) Time invariance

(D) None of these

Q.14 For the input x(t), an ideal impulse sampling system produces the output, $y(t) = \sum_{K=-\infty}^{0} x(kT)\,\delta(t - kT)$ where □(t) is the Dirac-delta function. The system is-

(A) Non-linear and time variant

(B) Non-linear and time invariant

(C) Linear and time variant

(D) Linear and time invariant

Q.15 Consider a system with input x(t) and output y(t) related as follows $y(t) = \frac{d}{dt}\{e^{-t}x(t)\}$ which of the following statement is true?

(A) System is stable

(B) System is time invariant

(C) System is non-linear

(D) System has memory

Q.16 An excitation is applied to a system at t = T and its response is zero for – □< t < T. Such a system is a

(A) Unstable system

(B) Stable system

(C) Causal system

(D) Non-casual system

Q.17 Consider a signal input single output discrete time system with x[n] as input and y[n] as output, where the two is related as-

$$y[n]\begin{cases} x[n]; & \text{for } 0 \le n \le 10 \\ x[n] - x[n-1]; & \text{otherwise} \end{cases}$$

System is-

(A) Causal and unstable

(B) Causal and stable

(C) Non-causal and unstable

(D) Non-causal and stable

Q.18 Determine the energy and power of the signal, MSQ

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

Choose the correct options from the following which is/are corrects.

(A) Energy = 4/3 J

(B) Power = 0 W

(C) Power = ∞ W

(D) Energy = 2/3 J

Q.19 For the following input output relationship determine whether the corresponding system is linear, time invariant (or) both.

$y(n) = x^2(n - 2)$

Choose the option from the following which is/are corrects?

(A) Non-linear, time invariant

(B) Linear, Time variant

(C) Non-linear, time variant

(D) Linear, time invariant

Q.20 Given h[n] = {2, 1, 2}. Let then by two system f[n] and g[n]. f[n] is obtained by convoluting h[n] by itself and g[n] is obtained by correlation of h[n]. Which one of the following statement is true?

(A) f[n] and g[n] has maximum value at origin

(B) f[n] has maximum value at f[0]

(C) g[n] has maximum value at g[0]

(D) None of the above

Q.21 The output relation of a system is given as y(t) = x(t) * h(t). An output g(t) is defined as g(t) = x(5t) * h(5t). The relation between y(t) and g(t) is given by:

g(t) = By(At)

Find A and B

(A) 5, 5 (B) $\frac{1}{5}, 5$

(C) $5, \frac{1}{5}$ (D) $\frac{1}{5}, \frac{1}{5}$

Q.22 The input signal $x(t) = e^{-2t}\,u(t)$ is applied to a system h(t) = u(t – 2). Find the region where output y(t) exist.

(A) t > 0 (B) 0 < t < 2

(C) t > 2 (D) t < 2

Q.23 Let the input to a system be defined as $x(t) = e^{-2t}u(t)$. Given, h(t) = u(t–2).

Find y(t) = x(t) * h(t)

(A) $\frac{1-e^{-2t}}{2};\ t > 0$

(B) $\frac{1-e^{-2t}}{2};\ t > 2$

(C) $\frac{1-e^{-2(t-2)}}{2};\ t > 2$

(D) $\frac{1-e^{-2(t-2)}}{2};\ t > 0$

Q.24 Consider a discrete-time LTI system with impulse response

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

Determine the response of the input signal

$$x(n) = \left(\frac{3}{4}\right)^n u(n)$$

(A) $-2\left(\frac{1}{2}\right)^n u(n) + 3\left(\frac{3}{4}\right)^n u(n)$

(B) $2\left(\frac{1}{2}\right)^n u(n) - 3\left(\frac{3}{4}\right)^n u(n);$

(C) $-2\left(\frac{1}{2}\right)^n u(n) - 3\left(\frac{3}{4}\right)^n u(-n)$

(D) $-2\left(\frac{1}{2}\right)^n u(n) + 3\left(\frac{3}{4}\right)^n u(n)$

Q.25 Consider a causal LTI system described by the difference equation,

$$y(n) + \frac{1}{2}y(n-1) = x(n)$$

Determine the frequency response $H(e^{j\omega})$ of this system. Also find response for the input $x(n) = \left(\frac{1}{2}\right)^n u(n)$.

(A) $H(e^{j\omega}) = \dfrac{1}{-1-\frac{1}{2}e^{-j\omega}}$ and

$$y(n) = \frac{1}{2}\left(-\frac{1}{2}\right)^n u(n) + \frac{1}{2}\left(\frac{1}{2}\right)^n u(n)$$

(B) $H(e^{j\omega}) = \dfrac{1}{-1-\frac{1}{2}e^{-j\omega}}$ and

$$y(n) = \frac{1}{2}\left(-\frac{1}{2}\right)^n u(-n) + \frac{1}{2}\left(\frac{1}{2}\right)^n u(-n)$$

(C) $H(e^{j\omega}) = \dfrac{1}{-1+\frac{1}{3}e^{-j\omega}}$ and

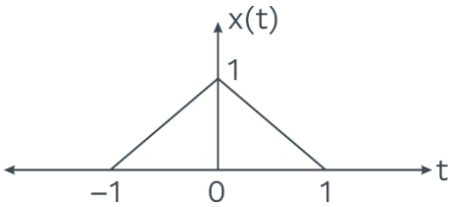
$$y(n) = \frac{1}{2}\left(\frac{1}{2}\right)^n u(-n) - \frac{1}{2}\left(\frac{1}{2}\right)^n u(-n)$$

(D) $H(e^{j\omega}) = \dfrac{1}{-1-\frac{1}{3}e^{-j\omega}}$ and

$$y(n) = \frac{1}{2}\left(\frac{1}{2}\right)^n u(n) - \frac{1}{2}\left(\frac{1}{2}\right)^n u(n)$$

Q.26 An input signal x(t) as shown in the figure is applied to a system whose impulse response is

$$h(t) = \sum_{K=-\infty}^{\infty} \delta(t-2k)$$



(1) Output signal is periodic

(2) Output signal is non periodic

(3) Output y(t) = x(t−2k)

(4) Output $y(t) = \sum_{K=-\infty}^{\infty} x(t-2k)$

Which of the following statement is correct?

(A) 1 only (B) 2 only

(C) 2 and 3 (D) 1 and 4

Q.27 Find the value of y(t) at t=3 which is defined as $\int h(t)$. The system is given by h(t) = δ(t−2) + □(t−5).

(A) 0 (B) 1

(C) 2 (D) infinity

3 Continuous time Fourier series

Q.1 Which of the following term does not have a Fourier Series representation?

(A) x(t) = Cos 5t + Sin 3t

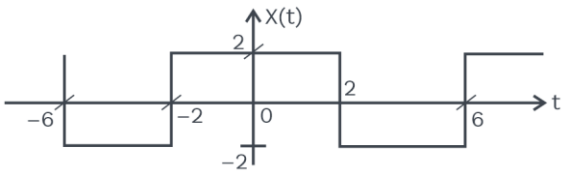
(B) x(t) = Cos 12t + Sin 16t + e^{j3t}

(C) $x(t)=\text{Sin}\left(\frac{\pi t}{8}\right)+\text{Sin}\left(\frac{\pi t}{3}\right)$

(D) None of these

(E) All of these

Q.2 The DC coefficient (a_0) of the CTFS of signal X(t) is shown in figure is -



(A) 2 (B) 16

(C) 0 (D) −8

Q.3 If f(t) = f(−t) & f(t) Satisfy the Dirichlet's conditions, then f(t) can be expanded in a Fourier Series containing -

(A) Only sine terms

(B) Only cosine terms

(C) Cosine terms & constant terms

(D) Sine terms and a constant term

Q.4 About the Fourier Series expansion of a periodic function it can be said that -

(A) Even function have only a constant & cosine term

(B) Odd function has only sine terms

(C) Even half wave symmetry contains only even harmonics with $a_0 = 0$.

(D) All of the above

(E) None of the above

Q.5 A periodic function of half wave symmetry is necessarily -

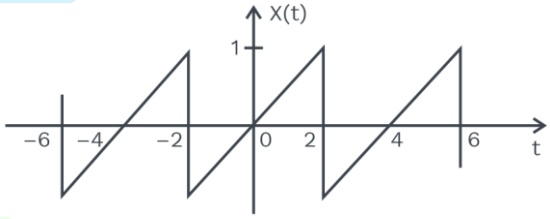
(A) An even function

(B) An odd function

(C) both even & odd

(D) Neither even nor odd

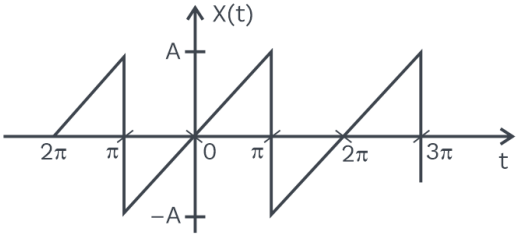
Q.6 Consider a periodic signal x(t) shown in figure. The trigonometric Fourier Series coefficient a_n is –



(A) 0 (B) $4\,\text{Sin}(\frac{\pi}{6}t)$

(C) $8\,\text{Sin}(\frac{\pi}{12}t)$ (D) $6\,\text{Cos}(\frac{\pi}{12}t)$

Q.7 The trigonometric Fourier Series of the periodic signal shown below –



(A) $\frac{A}{2}+\frac{4A}{\pi}\left(\text{Sin}\,t+\frac{1}{2}\text{Sin}\,2t+\frac{1}{3}\text{Sin}\,3t+\ldots\right)$

(B) $\frac{A}{2}+\frac{4A}{\pi}\left(\text{Cos}\,t+\frac{1}{3}\text{Cos}\,3t+\frac{1}{5}\text{Cos}\,5t+\ldots\right)$

(C) $\frac{4A}{\pi}\left(\text{Sin}\,t+\frac{1}{3}\text{Sin}\,3t+\frac{1}{5}\text{Sin}\,5t+\ldots\right)$

(D) $\frac{4A}{\pi}\left(\text{Cos}\,t+\frac{1}{2}\text{Cos}\,2t+\frac{1}{3}\text{Cos}\,3t+\ldots\right)$

Q.8 The Fourier series coefficient for the periodic signal x(t) = $\cos^2 t$ is -

(A) $\frac{-1}{4}\delta(n-1)+\frac{1}{2}\delta(n)-\frac{1}{4}\delta(n+1)$

(B) $\frac{1}{4}\delta(n-1)+\frac{1}{4}\delta(n)+\frac{1}{4}\delta(n+1)$

(C) $\frac{1}{4}\delta(n-1)+\frac{1}{2}\delta(n)+\frac{1}{4}\delta(n+1)$

(D) $\frac{1}{2}\delta(n-1)+\frac{1}{2}\delta(n)+\frac{1}{2}\delta(n+1)$

Q.9 A continuous time periodic signal has a fundamental period T = 10. The Fourier series coefficient which are non zero are –

$C_1 = C_{-1}^* = j/3$, $C_5 = C_{-5} = 6$

The signal will be-

(A) $\frac{2}{3}\sin\frac{\pi}{5}t+12\sin\frac{\pi}{5}t$

(B) $\frac{2}{3}\sin\frac{\pi}{5}t+12\cos\frac{\pi}{5}t$

(C) $-\frac{2}{3}\sin\frac{\pi}{5}t+12\cos\frac{\pi}{5}t$

(D) $\frac{2}{3}\sin\frac{\pi}{5}t-12\cos\frac{\pi}{5}t$

Q.10 The Fourier series co-efficient of signal is –

$C_0 = 4/\pi, c_1 = -j$

$C_n = \frac{4}{\pi(1-n^2)}$ for n even

Find Fourier series coefficient C_0' C_1' C_n' of signal when y (t) = x(t – 1)

(A) $C_0' = \frac{4}{\pi}, C_1' = j, C_n' = -\frac{4}{\pi(1-n^2)}$ n is odd

(B) $C_0' = 1/\pi, C_1' = -j, C_n' = -\frac{4}{\pi(1-n^2)}$ n is even

(C) $C_0' = 4/\pi, C_1' = -j, C_n' = \frac{4}{\pi(1-n^2)}$ n is even

(D) $C_0' = 4/\pi, C_1' = j, C_n' = \frac{4}{\pi(1-n^2)}$ n is odd

Q.11 A periodic voltage having Fourier series

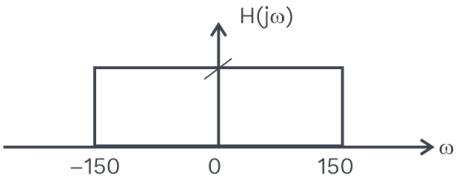
V(t) = 2 + 6 sin ωt + 12 cos ωt Volt

Is applied across a 4 Ω resistor the power dissipated in 4-ohm resistor is -

(A) 25 Watt (B) 20 Watt

(C) 30 Watt (D) 1 Watt

Q.12 A continuous time ideal low pass filter has the frequency response H(jω) shown in the figure



When an input x(t) with fundamental period, $T = 3\pi/5$ and Fourier series co-efficient, Cn is applied to the system, the output y(t) is same as x(t). for what value of n it is guaranteed that Cn = 0?

(A) |n| > 40 (B) |n| > 45

(C) |n| > 30 (D) |n| > 35

Q13. Match the List

P. f(t) = f(-t)	1. Convolution Integral

Q. $\int_0^t f_1(\tau) f_2(t-\tau)\,d\tau$	2.	Exponential form of Fourier series
R. $\sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$	3.	Laplace transform
S. $\int_{-\infty}^{\infty} f(t) e^{-st} dt$	4.	Even function wave symmetry
	5.	Fourier transform

(A) P → 4, Q →1, R → 2, S → 3

(B) P → 5, Q →2, R → 4, S → 3

(C) P → 5, Q →5, R → 3, S → 4

(D) P → 1, Q →2, R → 3, S → 4

Q.14 Let x(t) be a periodic signal with period T & Fourier series co-efficient c_n. Let y (t) = x (t – t_0) + x (t + t_0). The Fourier series co-efficient of y(t) is d_n. d_n = 0. For odd values of n. Then the value of t_0 is _____?

(A) T (B) T/2

(C) T/4 (D) T/8

Q.15 Find the Fourier series coefficient for the following signal –

X (t) = 6 cos (12t) + 4 sin (18t)

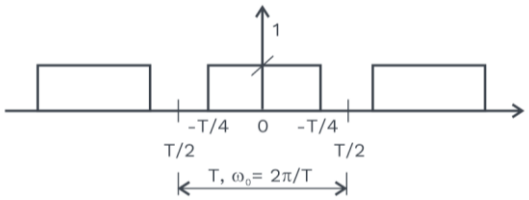
(A) a_0 = 0, a_1 = 4, b_1 = 0

(B) a_0 = 2, a_2 = 3, b_3 = 4

(C) a_0 = 0, a_2 = 6, b_3 = 4

(D) a_0 = 1/2, a_1 = 3, b_1 = 6

Q.16 Find the Trigonometry Fourier Series coefficient a_n of following signal. For even-



(A) $\frac{A\,Sin(n\pi/2)}{n\pi/2}$ (B) 0

(C) $\frac{A\,Cos(n\pi/2)}{n\pi/2}$ (D) $\frac{A\,Sin(n\pi/4)}{n\pi/4}$

Q.17 In the given Fourier Series

x(t) = 2 Sin (20πt + π/4) + 6 Cos (30πt + π/6)

Find the power present in the freq. range of 12 to 20Hz is _____

(A) 12 (B) 18

(C) 20 (D) 22

Q.18 x(t) is a periodic signal whose fundamental period. T = 8. The Fourier series coefficient are $C_a = C_{-1} = 2$ $C_3 = C_{-3}^* = \frac{8}{j}$ Find TFS is _____

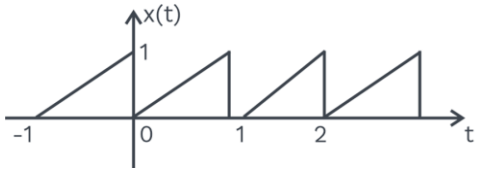
(A) $4\,Cos\frac{\pi}{4}t + 16\,Sin\frac{3\pi}{4}t$

(B) $6\,Sin\frac{\pi}{4} + 2\,Cos\frac{3\pi}{4}t$

(C) $16\,Cos\frac{\pi}{4}t + 4\,Cos\frac{3\pi}{4}t$

(D) $10\,Cos\frac{\pi}{3}t + 25\,Sin\,4\pi t$

Q.19 For the given exponential periodic signal find the exponential Fourier Series co-efficient.



(A) $\frac{i}{2\pi n}$ (B) $\frac{-i}{2\pi n}$

(C) $\frac{in}{2\pi}$ (D) $\frac{-in}{2\pi}$

Q.20 The Fourier series representation of periodic signal is $x(t)=\sum_{n=-\infty}^{+\infty}\frac{6}{2+(n\pi)^2}e^{jn\pi t}$ if one of the components is $A\,Cos\,3\pi t$. Find a_n

(A) 0 (B) $12/(4+9\pi^2)$

(C) $3/(4+3\pi^2)$ (D) $6/(4+\pi^2)$

Q.21 Find the power up to II harmonic for the given signal-



(A) 1 (B) 0.5

(C) 2 (D) 3

Q.22 The rms value of the resultant current in a wire which carries a dc current of 20A and a sinusoidal alternating current of peak value 30 A is -

(A) 14.1 A (B) 22.4 A

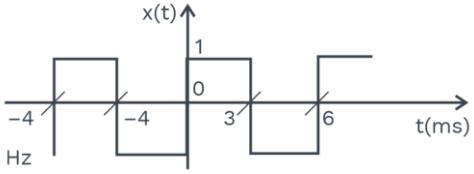
(C) 29.15 A (D) 30.0 A

Q.23 The period of the signal X(t) = 8, $Sin\left(0.6\pi t+\frac{\pi}{4}\right)$ is -

(A) 2.5π s (B) 3.33 s

(C) 4.4 s (D) 1.0π s

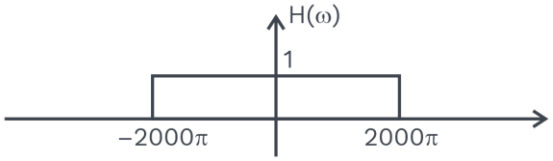
Q.24 A periodic rectangular signal has the waveform as shown in the figure. Frequency of the 9th harmonic of its spectrum



(A) 750 Hz (B) 1000 Hz

(C) 1250 Hz (D) 1500 Hz

Q.25 Let the signal $x(t)=\sum_{K}^{+\infty}(-1)^K\delta(t-\frac{K}{1000})$ be passed out through an L.T.I system with frequency response H(ω) as given in the figure below:



Find the output when signal is passed through L.T.I system-

(A) 2000 Cos (1000πt)

(B) 2000 + 2000 Cos (2000πt)

(C) 2000 Sin (2000πt)

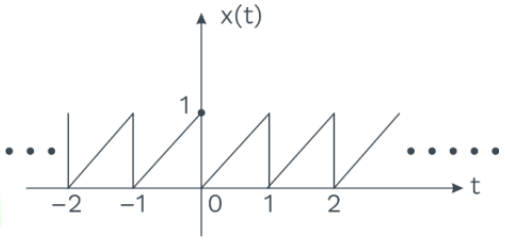
(D) 1000 + 1000 Cos (1000πt)

Q.26 Current is flowing in a wire which has dc component of 5A and ac component of 4A. Calculate the RMS value of the resultant current:

(A) 3 A (B) 5.74 A

(C) 6.36A (D) 4.52 A

Q.27 The input signal x(t) is given as:



Consider the statement below regarding input signal x(t) and state which of the following statement is TRUE?

(A) Odd symmetry

(B) Odd half wave symmetry

(C) DC component and odd symmetry

(D) DC component and odd half wave symmetry

4 Discrete time Fourier Series

Q.1 Considers a signal x(n) with the following data that x(n) is real and even signal. The period of x(n) is N = 12. The value of $C_{13}=6$. It is given that $\frac{1}{12}\sum_{n=0}^{11}|C_k|^2=72$. So find the signal x(n) is ___

(A) $12\sin\frac{\pi}{6}n$ (B) $12\cos\pi/6n$

(C) $12j\sin\frac{\pi}{12}n$ (D) $12j\sin\frac{\pi}{6}n$

Q.2 x[n] is real and odd periodic signal with period of N = 9 and Fourier coefficient a_k.

Given that $a_{28}=24j$ then a_{-1}/j^3 is ____

(A) 24 (B) 24/j

(C) 24j (D) -24

Q.3 A discrete time signal is given as $x[n]=\cos\left(\frac{8\pi}{21}n\right)+\cos\left(\frac{22\pi}{41}n\right)+14$. If C_k is the Fourier series coefficient of x[n], the jc_{10} is

(A) ½ (B) -1/2

(C) $\frac{1}{2}j$ (D) $\frac{1}{2j}$

Q.4 Discrete time Fourier series expansion of $x[n]=\sin(1.2\pi n)\sin(0.5\pi n)$ will contain the harmonics.

(A) 17th, 13th, 7th, 3rd

(B) 12th, 11th, 7th, 3rd

(C) 17th, 13th, 7th, 2nd

(D) 17th, 13th, 2nd, 5th

Q.5 The DTFS coefficient of signal x[n] with fundamental period N = 6 are given as

$C_1=4+j6,\ C_0=6,\ C_1=4-j6,\ c_2=5$

The co-efficient $j\,c_{25}$ is

(A) 4 – j6 (B) 6 – j4

(C) 6 + j4 (D) 4 + j6

Q.6 The DTFS coefficient of a signal x[n] is given below

$c_k=2j\sin\left(\frac{6\pi}{23}k\right)+\sin\left(\frac{20\pi}{23}k\right)$

The signal x[n] is___

(A) $23[\delta(n-3)-\delta(n+3)]+\frac{23}{2j}[8(n-10)-\delta(n+10)]$

(B) $[\delta(n-3)-\delta[n+3]]+\frac{1}{2j}[\delta(n-10)-\delta(n+10)]$

(C) $23[\delta[n+3]-\delta(n-3)]+\frac{23}{2j}[\delta(n+10)-\delta(n-10)]$

(D) $\left[\delta[n+3]-\delta[n-3]+\frac{1}{Qi}[\delta(n+10)-\delta(n-10)]\right.$

Q.7 What is the discrete time Fourier coefficients of $\sum_{m=-\infty}^{\infty}\delta(n-3m)$ for all k?

(A) 1/3 (B) 3

(C) 1 (D) $\sqrt{3}$

Q.8 x[n] be the periodic extension of the sequence $\delta[n]-\delta[n-3]$ with period 5. The DTFS co-efficient of $x[n]$ is $\frac{1}{A}\left[1-e^{-jB\pi k}\right]$. Find the value of A & B

(A) A = 1/5, B = 6/5

(B) A = 5, B = 3/5

(C) A = 5, B = 6/5

(D) A = 1/5, B = 3/5

Q.9 The DTFS coefficient of a signal x[n] is $C_k = \sin\left(\frac{10\pi k}{13}\right)$ the signal x[n] is-

(A) $\frac{13}{2j}[\delta[n+5]-\delta[n-5]] \quad |n| \le 7$

(B) $13\,[\delta[n+5]-\delta[n-5]] |n| \le 7$

(C) $\frac{13}{2j}[\delta(n-5)-\delta[n+5]] |n| \le 7$

(D) $13\ \sum \delta[n-5]+\delta[n+5]\big] |n|\big) \le 7$

Q.10 Discrete time Fourier series does not exist for the signal?

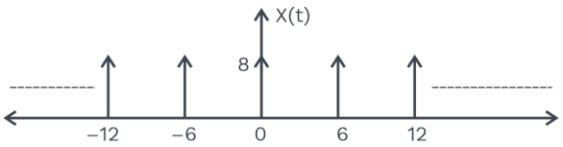
(A) $x(n) = \cos(3\sqrt{2}\pi n)$

(B) $x(n) = \sin(\pi n/5)$

(C) $x(n) = \sum_{k=-\infty}^{\infty} \delta(n-12k)$

(D) Both A & C

Q.11 What is the Fourier Series coefficient for the periodic signal shown below -



(A) $C_n = 3$ (B) $C_n = 4$

(C) $C_n = 4/3$ (D) $C_n = ¾$

5 Fourier Transforms

Q.1 Fourier transform is valid for

(A) Periodic signal

(B) non-periodic signal

(C) a & b

(D) None

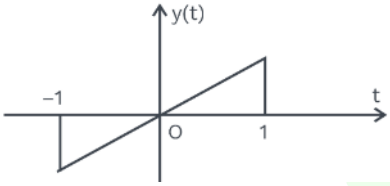
Q.2 Find the Fourier transform of –

$y(t) = t \cdot e^{-4t} u(t)$

(A) $\frac{1}{(4+j\omega)^2}$ (B) $\frac{1}{4+j\omega}$

(C) $\frac{4}{(4+j\omega)^2}$ (D) $\frac{4}{\left(\frac{1+j\omega}{4}\right)^2}$

Q.3 Given that $x(t) \rightarrow X(\omega) = \frac{4\sin\omega}{\omega}$. Find the FT of the following signals in terms of X(w)-



(A) $\frac{d}{d\omega}\left[\frac{4\sin\omega}{\omega}\right]$ (B) $\frac{j4\sin\omega}{\omega}$

(C) $\frac{d}{d\omega}\left(\frac{4\sin\omega}{\omega}\right)$ (D) $\frac{4\sin\omega}{\omega}$

Q.4 A LTI system whose Impulse response $h(t) = \frac{\sin 2t}{\pi t}$ is applied with an input x(t) = 2cost+10sin8t find the o/p of the LTI system

(A) 2 cos t

(B) 10 sin 8t

(C) 2 cos t + 10 Sin 8t

(D) None

Q.5 Find the Fourier transform of the following

$y(t) = \frac{\text{Sin}2t \,.\, \text{Sin}t}{\pi t^2}$

(A) ω + 3 (B) ω

(C) ω - 3 (D) 0

Q.6 Find the value of $\int_{-\infty}^{t} \frac{\text{Sin}4\tau}{\pi\tau} d\tau$

(A) $\frac{\text{rect}(\omega/8)}{j\omega}$

(B) $\frac{\text{rect}(\omega/4)}{j\omega}$

(C) $\frac{\text{rect}(\omega/8)}{j\omega} + \pi d\omega$

(D) $\frac{\text{rect}(\omega/4)}{j\omega} + 2\pi\delta(\omega)$

Q.7 Find the energy of $y(t) = \frac{2\text{Sin}4\pi t}{\pi t}\text{Cos}2\pi t$

(A) 4J (B) 6 J

(C) 16 J (D) 36 J

Q.8 Match the time domain signal to its equivalent frequency domain to its equivalent frequency domain representation.

List I	List II
(i) $x(t)e^{-a\|t\|}$	(a) -2π sgn(ω)
(ii) sgn(t)	(b) $\frac{2a}{a^2+\omega^2}$
(iii) $\frac{2}{jt}$	(c) $\frac{2}{j\omega}$

(A) i → a, ii → b, iii → c

(B) i → c, ii → b, iii → a

(C) i → a, ii → c, iii → b

(D) i → b, ii → c, iii → a

Q.9 Match the frequency domain signal to time domain signal. Fourier transform

(i) $e^{-jt0\omega}$ (a) Sin $\omega_0 t$

(ii) $\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$ (b) cos $\omega_0 t$

(iii) $\frac{\pi}{j}\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$ (c) $\delta(t-t_0)$

(A) i → c, ii → a, iii → b

(B) i → a, ii → b, iii → c

(C) i → b, ii → c, iii → a

(D) i → c, ii → b, iii → a

Q.10 Fourier transform of $\frac{4t}{(1+t^2)^2}$ is

(A) $-j2\pi\omega e^{-|\omega|}$ (B) $-j4\pi^2 f e^{-|\omega|}$

(C both a & b (D) $j2\pi\omega e^{-|\omega|}$

Q.11 Find the value of a, If Z (ω) = aω

(A) j (B) 2

(C) 2j (D) 2jω

Q.12 Find the value of B, if $B=\int_{-\infty}^{\infty}|k(t)|^2\,dt$

(A) $\frac{4}{5\pi}$ (B) $\frac{1}{5\pi}$

(C) $\frac{4\pi}{5}$ (D) $\frac{\pi}{5}$

Q.13



Let h (t) = 1 If x(t) = ..., then y(t) is given by –

(A) Sin t (B) Cos t

(C) -Sin t (D) -Cos t

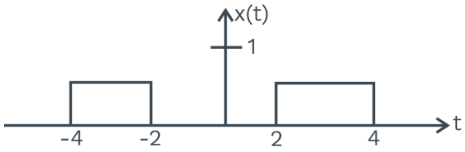
Q.14 A signal $x(t)=2e^{-\pi t^2}$ is convolved with $h(t)=\left(e^{-\pi t^2/2}\right)$. If the resultant signal is y(t) –

The energy of the signal x(t) is -

(A) $4\sqrt{2}$ (B) $2\sqrt{2}$

(C) $8\sqrt{2}$ (D) 8

Q.15 Find the Fourier transform of the given function –



(A) 2w(cos2w - sin4w)

(B) 2w(cos2w - cos4w)

(C) sin2w - sin4w

(D) 2w(sin2w - 2wsin4w)

Q.16 A real and non-negative signal x(t) has Fourier transform (jw) suppose we are given the following facts –

(1) $FT^{-1}\{(4+j\omega)X(\omega)\}=A\,e^{-3t}\,u(t)$

(2) $\int_{-\infty}^{\infty}|X(j\omega)|^2\,d\omega=8\pi$

Where A is constant –

The signal x(t) may be

(i) $2/5\sqrt{84}\left(e^{-4t}-e^{-3t}\right)u(t)$

(ii) $-\frac{2}{5}\sqrt{84}\left(e^{-4t}-e^{-3t}\right)u(t)$

(A) only (B) (ii) only

(C) both (D) None

Q.17 What is the energy of the signal x(t) = e^{-6t} u(t) in the frequency band |w| ≤ 1.5rad/sec?

(A) 0.013 (B) 0.743

(C) 0.546 (D) 0.104

Q.18 The inverse fourier transform of $X(j\omega) = \frac{16}{\omega^2 + 64}$ will be stable if x(t) is –

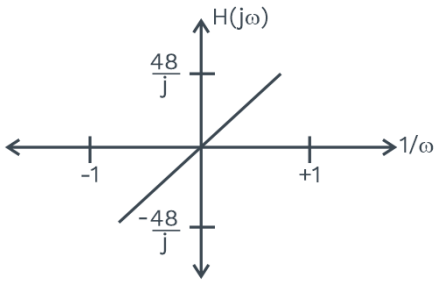
(A) $e^{-8t}u(t) + e^{8t}u(-t)$

(B) $-e^{-8t}u(-t) - e^{8t}u(t)$

(C) $e^{-8t}u(t) + e^{8t}u(-t)$

(D) $-e^{-8t}u(-t) + e^{8t}u(t)$

Q.19 A causal LTI filter has the frequency response H(jw) given in figure. For input signal $x(t) = e^{-j24t}$, output will be –

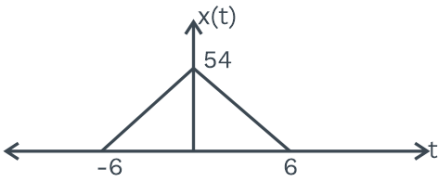


(A) $-2je^{-j24t}$ (B) 0

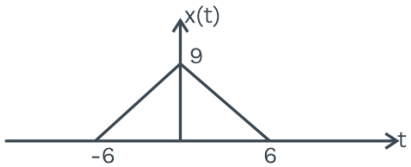
(C) $2je^{-j24t}$ (D) je^{-j24t}

Q.20 If $x(t) \xleftrightarrow{FT} X(j\omega) = 54\sin C^2(3\omega/\pi)$ then which of the following corresponds to signal x(t)?

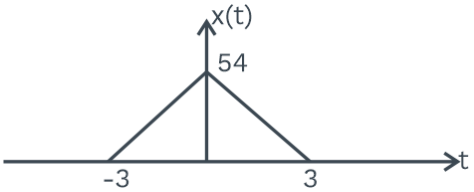
(A)



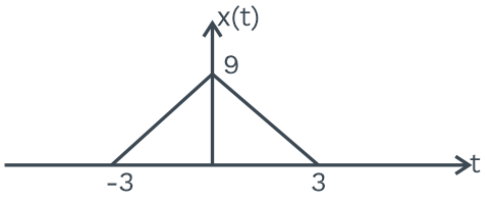
(B)



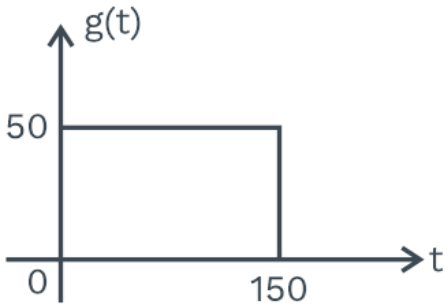
(C)



(D)



Q.21 The energy density spectrum G(f) of a rectangular pulse shown in the figure is –



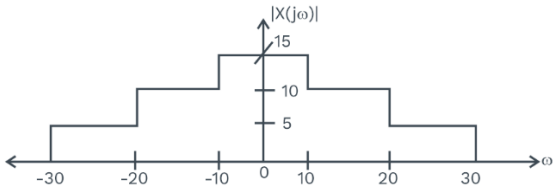
(A) $150^2\left(\frac{\sin(150\pi f)}{150\pi f}\right)^2$

(B) $50^2\left(\frac{\sin(150\pi f)}{150\pi f}\right)^2$

(C) $150 \times 50\frac{\sin(150\pi f)}{(150\pi f)}$

(D) $(150 \times 50)^2\left[\frac{\sin(150\pi f)}{150\pi f}\right]^2$

Q.22 The energy of a signal whose Fourier transform is shown in figure equals to –



(A) 1590J (B) 1590W

(C) 1600J (D) 1600W

Q.23 What is the total area under the signal x(t)?

(A) 0 (B) 48

(C) 72 (D) 3/2

Q.24 If $y(t) = \int_{-\infty}^{\tau} x(\tau)d\tau$ then the total area under y(t) is

(A) 0 (B) 0.02

(C) 1.5 (D) 0.67

Q.25 Which of the term is not true for Laplace transform.

(a) Generalization of FT is Laplace transform.

(b) Laplace is defined for stable system only.

(c) FT is special case of Laplace transform where a = 0

(d) Laplace is used for investigation of stability of FB.

(A) c (B) d

(C) b (D) a

Q.26 If G(f) represents the Fourier transform of a signal g(t) which is real and odd symmetric in time, then:

(A) G(f) is complex

(B) G(f) is imaginary

(C) G(f) is real

(D) G(f) is real and non-negative

Q.27 Let $x[n] = \left(\frac{1}{2}\right)^n u(n), y[n] = x^2[n]$ and $Y[e^{j\Omega}]$ be the Fourier transform of y[n]. Then $Y[e^{j0}]$ is: _____ (up to 2 decimal places)

Q.28 Property of the Fourier transform which states that the compression in time domain is equivalent to the expansion in the frequency domain is:

(A) duality

(B) frequency shifting

(C) scaling

(D) time scaling

Q.29 Find the inverse Fourier transform of $u(\omega)$

(A) $\frac{1}{2}\delta(t) + \frac{1}{2\pi}\frac{j}{t}$ (B) $\delta(t) + \frac{1}{\pi}\frac{j}{t}$

(C) $\delta(t) - \frac{1}{2\pi}\frac{j}{t}$ (D) $\frac{1}{2}\delta(t) - \frac{1}{2\pi}\frac{j}{t}$

Q.30 The discrete sequence is given as x[n] = {1, 2, 3, 4}. Which of the following relation holds true?

(A) X[3] = X*[1], X[2] = -2

(B) X[3] = –X[1], X[2] = -2

(C) X[1] = –X[3], X[2] = 2

(D) X[2] = X*[1], X[2] = 2

Q.31 The input to a channel is a bandpass signal which is linearly modulated with a carrier with single tone signal

$y(t) = \cos(100t - 1.56)\cos(10^6t - 10^{-6})$

Which of the following statement holds true?

(A) $t_{p(\omega)}$ = 1.56, phase distortion

(B) $t_{p(\omega)}$ = 1.56×10^{-2}, no phase distortion

(C) $t_{g(\omega)}$ = 1.56, no phase distortion

(D) $t_{g(\omega)}$ = 1.56×10^{-2}, phase distortion

Q.32 Find the energy of the signal

$$y(t) = 4\frac{\sin 4\pi t}{\pi t}\cos 2\pi t$$

E = _____ Joules.

Q.33 The input and output of causal LTI sys is related by the differential

equation $\frac{d^2y(t)}{dt^2}+\frac{5dy(t)}{dt}+6y(t)=x(t)$.

Find the response of the system if $x(t)=e^{-4t}u(t)$.

(A) $\left(e^{4t}+e^{3t}+e^{2t}\right)u(t)$

(B) $\frac{1}{2}\left[e^{4t}+2e^{+3t}+e^{+2t}\right]u(t)$

(C) $\frac{1}{2}\left[e^{-4t}-2e^{-3t}+e^{-2t}\right]u(t)$

(D) $\left[e^{-4t}-2e^{-3t}+e^{-2t}\right]u(t)$

Q.34 The 3-dB bandwidth of a low pass signal $e^{-t}u(t)$ is the unit step function, is given by.

(A) $\frac{1}{\pi}$Hz (B) $\frac{1}{2\pi}$Hz

(C) 1 Hz (D) $\frac{1}{2\pi}\sqrt{\sqrt{2}-1}$Hz

Q.35 The energy of the signal $x(t)=\frac{\sin 4\pi t}{4\pi t}$ is J.

Q.36 Let $g(t)=e^{-\pi t^2}$ and h(t) is a filter matched to g(t). If g(t) is applied as input to h(t), then the Fourier transform of the output is.

(A) $e^{-2\pi f^2}$ (B) $e^{-\pi f^2/2}$

(C) $e^{-\pi f^2}$ (D) $e^{\pi f^2}$

Q.37 Match the following and choose the correct combination.

W: Continuous and aperiodic signal.

X: Continuous and periodic signal.

Y: Discrete and aperiodic signal.

Z: Discrete and periodic signal.

1. Fourier representation is continuous and aperiodic.
2. Fourier representation is discrete and aperiodic.
3. Fourier representation is continuous and periodic.
4. Fourier representation is discrete and periodic.

(A) W–1, X–2, Y–3, Z–4

(B) W–2, X–1, Y–4, Z–3

(C) W–1, X–3, Y–2, Z–4

(D) W–3, X–2, Y–4, Z–1

Q.38 A unit step input is applied to a system which is defined as h(t)=u(t). Find the output of the system, y(t).

(A) u(t) (B) t u(t)

(C) $\frac{t^2}{2}u(t)$ (D) $\delta(t)$

Q.39 The Fourier transform of a function g(t) is given as

$$G(\omega)=\frac{\omega^2+24}{\omega^2+4}$$

(A) sin 2ωt + 5 cos ωt

(B) $\delta(t)$ - $5e^{2|t|}$

(C) $\delta(t)$ + $5e^{-2|t|}$

(D) cos 2ωt + $20e^{-2t}$

Q.40 Determine the Fourier transform of the following signals:

$$\left(\frac{1}{2}\right)^{n-1}u(n-1)$$

(A) $e^{-j\omega}\cdot\frac{2}{2-e^{-j\omega}}$ (B) $e^{j\omega}\cdot\frac{2}{2-e^{-j\omega}}$

(C) $e^{-j\omega}\cdot\frac{2}{2+e^{-j\omega}}$ (D) $e^{-j\omega}\cdot\frac{2}{2-\theta^{j\omega}}$

Q.41 Calculate the value of X(1), for the following signal,

$x(n)=\delta(n-1)+\delta(n+1)$

Q.42 Determine the inverse Fourier transform of

$$X\left(e^{j\omega}\right)=\begin{cases}2j, & 0<\omega\le\pi\\ -2j, & -\pi<\omega\le 0\end{cases}$$

(A) $\frac{2}{n\pi}[\cos n\pi+1]$

(B) $\frac{2}{n\pi}[\cos n\pi-1]$

(C) $-\frac{2}{n\pi}[\cos n\pi+1]$

(D) $-\frac{2}{n\pi}[\cos n\pi-1]$

Q.43 Compute the Fourier transform of the following signal.

$$x(n)=(2)^n\sin\left(\frac{\pi}{4}n\right)u(-n)$$

(A) $-\frac{1}{2j}\left[\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega+\frac{\pi}{4}\right)}}-\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega-\frac{\pi}{4}\right)}}\right]$

(B) $\frac{1}{2j}\left[\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega+\frac{\pi}{4}\right)}}+\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega-\frac{\pi}{4}\right)}}\right]$

(C) $-\frac{1}{j}\left[\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega+\frac{\pi}{4}\right)}}+\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega-\frac{\pi}{4}\right)}}\right]$

(D) $\frac{1}{j}\left[\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega+\frac{\pi}{4}\right)}}-\frac{1}{1-\left(\frac{1}{2}\right)e^{j\left(\omega-\frac{\pi}{4}\right)}}\right]$

Q.44 The Fourier transform of discrete time signal is given. Determine the signal corresponding to the transform.

$$X(e^{j\omega})=\cos^2\omega+\sin^2 3\omega$$

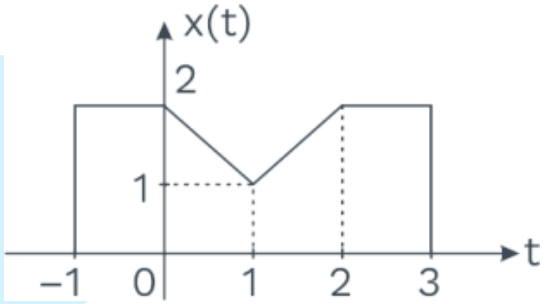
(A) $\delta(n)+\frac{1}{4}\delta(n-3)+\frac{1}{4}\delta(n+2)+\frac{1}{4}\delta(n+6)-\frac{1}{4}\delta(n-6)$

(B) $\delta(n)+\frac{1}{4}\delta(n-2)+\frac{1}{4}\delta(n+2)-\frac{1}{4}\delta(n+6)-\frac{1}{4}\delta(n-6)$

(C) $\delta(n)+\frac{1}{4}\delta(n+3)+\frac{1}{4}\delta(n+2)-\frac{1}{4}\delta(n+6)-\frac{1}{4}\delta(n-6)$

(D) $\delta(n)+\frac{1}{4}\delta(n-2)+\frac{1}{4}\delta(n+2)+\frac{1}{4}\delta(n+6)+\frac{1}{4}\delta(n-6)$

Q.45 Find the values of the following signal shown below:



Find the value of X(0) _____

Q.46 An LTI system is having impulse response $h(t)=\frac{\sin 4t}{\pi t}$ is excited with an input x(t) = cos2t + sin6t. Find the output y(t).

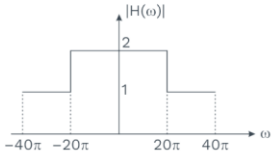
(A) cos2t

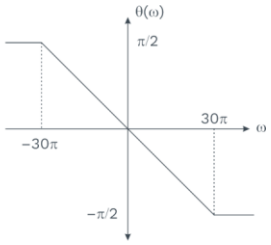
(B) sin6t

(C) cos2t + sin6t.

(D)$\cos^2 2t+\sin^2 6t$.

Q.47 Consider a transmission system shown in the figure with magnitude and phase response. Let the input be x(t) = 2cos(10πt) + sin(26πt). Consider the statement below and state which one is TRUE?





(A) phase distortion

(B) amplitude distortion

(C) phase and amplitude distortion

(D) distortion less

Q.48 The system equation is given as

$$h[n] = 4\sqrt{2}, \quad n = \pm 2$$
$$= 2\sqrt{2}, \quad n = \pm 1$$
$$= 0, \quad \text{elsewhere}$$

Find the value of $H[e^{j\Omega}]$ at $\omega=(\pi/4)$

Q.49 Find the value of $X(\omega) = \sum_{n=-\infty}^{\infty} (-1)^n x[n]$,

where $X(\omega) = \cos^3(3\omega)$

Q.50 Consider a system consisting of the cascade of two LT system with frequency responses.

$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

And,

$$H_2(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-j2\omega}}$$

Find the difference equation of the overall system.

(A) $y(n) + \frac{1}{8}y(n-3) = 2x(n) + x(n-1)$

(B) $y(n) + \frac{1}{8}y(n+2) = 2x(n) - x(n-1)$

(C) $y(n) + \frac{1}{8}y(n-3) = 2x(n) - x(n-1)$

(D) $y(n) - \frac{1}{8}y(n-3) = 2x(n) - x(n+1)$

Q.51 The discrete time sequence is given as x[n] = [3, 0, –1, –2, 4, 9, –5]. Find the value of $X(e^{j\pi})$ is:

(A) 6 (B) –6

(C) j(6) (D) –j(6)

Q.52 For the N-point FFT algorithm with $N = 2^m$, which one of the following statements is TRUE?

(A) It is not possible to construct a signal flow graph with input and output in normal order.

(B) The number of butterfly structure in m^{th} state is N/m

(C) Computation of butterfly requires only one complex multiplication.

(D) In place of computation it requires storage of only 2N node data.

6 Laplace Transforms

Q.1 Identify the transfer function corresponding to an all-pass filter from the following:

(A) $\frac{1-s\tau}{1+s\tau}$ (B) $\frac{s\tau}{1+s\tau}$

(C) $\frac{1+s\tau_1}{1+s\tau_2}$ (D) $\frac{1}{1+s\tau}$

Q.2 For the given signal $X(s)=\frac{2s+5}{s^2+5s+6}$ state the ROC for which the signal is non-causal and unstable.

(A) $\sigma > -2$

(B) $\sigma < -3$

(C) $\sigma > -2$ and $\sigma < -3$

(D) $-3 < \sigma < -2$

Q.3 The inverse Laplace transform of $Y(s)=\frac{e^{-3s}}{(s+1)(s+2)}$ is:

(A) $e^{t}\,u(t) - e^{2t}\,u(t)$

(B) $e^{-(t+3)}\,u(t+3) - e^{-2(t+3)}\,u(t+3)$

(C) $e^{-(t-3)}u(t-3) - e^{-2(t-3)}\,u(t-3)$

(D) $e^{-t}\,u(t) - e^{-2t}u(t)$

Q.4 The system equation is defined by

$$H(s)=\frac{s}{(s+1)(s+2)}$$

The input to the system is $x(t) = e^{2t}$ for all t. Find the output signal, y(t):

(A) e^{2t} (B) $\frac{1}{6}e^{2t}$

(C) e^{-2t} (D) $\frac{1}{6}e^{-2t}$

Q.5 Find the Initial value theorem & Final value theorem for the following signals –

$$X(s)=\frac{2S+5}{S^2+5S+6}$$

(A) IVT = 2, FVT = 0

(B) IVT = 0, FVT = 2

(C) IVT = 2, FVT = -4

(D) IVT = 0, FVT = 0

Q.6 Which are the properties of region of convergence of Laplace transform.

(A) ROC does not include the pole.

(B) ROC of a Laplace transform is the parallel to jω axis.

(C) If function is finite duration & absolute integrable then the ROC is entire S-plane.

(D) All of the Above.

Q.7 For what value of x(t) the Laplace transform is $\frac{1}{s+a}$

(i) $x(t) = e^{-at}\,u(t)$ $(a > 0)$

(ii) $x(t) = e^{-at}\,u(t)$ $(a < 0)$

(iii) $x(t) = -e^{-at}\,u(-t)$ $(a > 0)$

(iv) $x(t) = -e^{-at}\,u(-t)$ $(a < 0)$

(A) i (B) i, ii, iii

(C) i, iv (D) i, ii, iii, iv

Q.8 Which of the function has the ROC entire plane -

(A) $\delta(t)$ (B) $u(t)$

(C) $u(-t)$ (D) Both a & b

Q.9 For the given system

$$H(S) = \frac{S-2}{(S+3)(S-4)}$$

Which are the stable system in time domain -

(i) $h(t) = 2/3\ e^{-t}\ u(t) + 1/3\ e^{2t}u(t)$

(ii) $h(t) = \frac{-2}{3}e^{-t}u(-t) - \frac{1}{3}e^{-4t}u(-t)$

(iii) $h(t) = \frac{2}{3}e^{-t}u(-t) - \frac{1}{3}e^{2t}u(-t)$

(A) Only iii (B) Only ii

(C) i, ii (D) I, ii, iii

Q.10 Determine the Laplace transform for the following signal –

$y(t) = e^{-6t}\ u(t-3)$

$x(t) = e^{-6t}\ u(t)$

(A) $\frac{e^{-3(S-6)}}{S+6}$ (B) $\frac{e^{-3(S+6)}}{S+6}$

(C) $\frac{e^{-3(S-6)}}{S-6}$ (D) $\frac{e^{-(3S+6)}}{S+6}$

Q.11 Let x(t) be signal, that has rational Laplace transform. Its pole are located at

$S = -2, S = -4$

If $y(t) = e^{3t}x(t)$

Determine

the ROC for the stability.

(A) $-1 < ROC < 1$ (B) $-3 < ROC < 4$

(C) $-1 > ROC > 1$ (D) $4 < ROC < -3$

Q.12 Find the Inverse Laplace transform of $\log\left(\frac{S+4}{S+5}\right)$?

(A) $\frac{e^{-5t}+e^{-4t}}{t}u(t)$ (B) $\frac{e^{-5t}-e^{-4t}}{t}$

(C) $\frac{e^{-4t}-e^{-5t}}{t}$ (D) $\frac{e^{-5t}-e^{-4t}}{t}u(t)$

Q.13 Consider an LTI system whose response to input $x(t) = [e^{-3t} + e^{-5t}]4(t)$ is $y(t) = [2e^{-3t} - 2e^{-6t}]u(t)$. If the system has poles at S1 & S2 and zero at Z then they are respectively given by

(A) $S_1 = -6, S_2 = -4, Z = -5$

(B) $S_1 = 6, S_2 = -4, Z = -5$

(C) $S_1 = 6, S_2 = 4, Z = 5$

(D) $S_1 = -6, S_2 = 4, Z = -5$

Q.14 The laplace transform of $\delta(6t)$ & $u(10t)$

(A) $\left(\frac{1}{6}\right), 1$ (B) $1, \frac{1}{10}$

(C) $\frac{1}{6}, \frac{1}{10}$ (D) $\frac{1}{6}, \frac{1}{s}$

Q.15 The laplace transform of a continuous time signal x(t) is $X(s) = \frac{7-S}{S^2-S-30}$. If the Fourier transform of the signal exists then-

(A) $\frac{2}{12}e^{5t}u(-t)\frac{-13}{12}e^{-6t}u(t)$

(B) $\frac{-2}{12}e^{5t}u(-t) + \frac{13}{12}e^{-6t}u(t)$

(C) $\frac{-2}{12}e^{5t}u(-t) - \frac{13}{12}e^{-6t}u(t)$

(D) $\frac{-13}{12}e^{5t}u(-t) - \frac{2}{12}e^{-6t}u(t)$

Q.16 The laplace transform of a function f(t) is given by

$F(S) = \ln\left(\frac{S+3}{S+4}\right)$ then f(t) for $t > 0$ is

(A) $\frac{e^{-3t}-e^{-4t}}{t}$ (B) $\frac{e^{-4t}-e^{-3t}}{t}$

(C) $\frac{e^{-4t}+e^{-3t}}{t}$ (D) $\frac{e^{-3t}+e^{-4t}}{t}$

Q.17 If $F(S) = L[f(t)] = \frac{64(S+6)(S+13)}{(S+2)(S^2+4)}$ then $\lim_{t\to\infty} f(t)$ is given by

(A) 24 (B) 0

(C) infinite (D) undefined

Q.18 A. Assertion : A system with transfer function $Z(S) = \frac{1}{S-6}\ \sigma < 6$ is stable

R. Reason : The system is Non causal

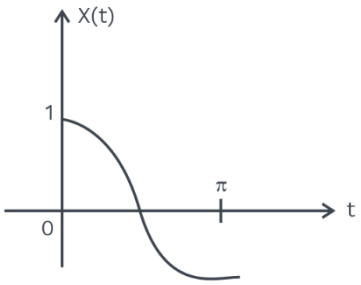
(A) both A & R are true and R is the correct explanation of A

(B) Both A & R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q.19 The laplace transform of the signal x(t) is X(s) than X(s) is



(A) $\frac{1+e^{-5\pi}}{1+S^2}$ (B) $\frac{S+e^{-5\pi}}{1+S^2}$

(C) $\frac{S(1+e^{-5\pi})}{1+S^2}$ (D) $\frac{1}{1+S^2}$

Q.20 If the laplace transform of the voltage across a capacitor of value 5 e is $V_c(s) = \frac{S+2}{S^3+2S^2+2S+4}$, the value of the current through the capacitor at t = 0+ is

(A) 0A (B) 5A

(C) 1/5 A (D) 1A

Q.21 $\frac{dh(t)}{dt} + 3h(t) = e^{-6t}u(t) + Bu(t)$ is the LTI system with the impulse response satisfies the differential equation given above.

Where b is unknown constant the impulse response has poles at -

(A) S = 0, S = -6, S = -3

(B) S = 0, S = -6

(C) S = 0, S = 6, S = 3

(D) S = -6, S = -3

Q.22 Find the value y' + 2y = 4t, y(0) = 3 in time domain.

(A) $4e^{-2t}$

(B) $4e^{-2t} - 1 + 2t$

(C) 0

(D) $4e^{-2t}u(t) - u(t) + 2t\,u(t)$

Q.23 Find the constant A, B, y_0 and y_0' so that $y(s) = \frac{2s-6}{4s^2+s+20}$ is the Laplace transform of the given problem:

$Y'' + Ay' + By = 0,\ y(0) = y_0$

$Y'(0) = y'_0$

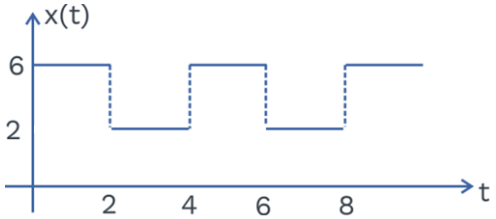
(A) A = 1, B = 20, $y_0^1 = -8$, $y_0 = 2$

(B) A = 20, B = 1, $y_0^1 = -8$

(C) B = 0, A = 20, $y_0^1 = 8$

(D) A = 0, B = 0, C = −8

Q.24 Find the Laplace transform of the periodic function.



(A) $\frac{6-4e^{-2s}-2e^{-4s}}{s(1-e^{2s})}$

(B) $\frac{6+4e^{-2s}-2e^{-4s}}{s}$

(C) $\frac{1}{s}(6+4e^{-2s}+2e^{-4s})$

(D) $\frac{1}{s}(6-4e^{-2s})$

Q.25 Determine the period of the x(t) and find its Laplace transform.

$$x(t)=\begin{cases}\sin t & 0\le t<\pi\\ 0 & \pi\le t<2\pi\end{cases}$$

$X(t + 2p) = x(t)\ t \geq 0$

(A) $2\pi, \frac{1+e^{-\pi s}}{(1+s^2)(1-e^{-2\pi s})}$

(B) $\pi, \frac{1+e^{-\pi s}}{(1+s^2)(1-e^{-2\pi s})}$

(C) $2\pi, \frac{1+e^{-\pi s}}{1-e^{-2\pi s}}$

(D) $2\pi, \frac{1+e^{-\pi s}}{1+s^2}$

Q.26 Find out the ROC and LT → $f(t) = e^{-3(t)}$ for which the system is stable:

(A) $-3 < a < 3; \frac{6}{s^2-9}$

(B) $-3 < a < 3; -\frac{6}{s^2-9}$

(C) $-3 > a > 3; \frac{6}{s^2-9}$

(D) $-3 > a > 3; -\frac{6}{s^2-9}$

Q.27 $x(t) \rightleftharpoons x(s)$

$$y(t)=\int_{-\infty}^{\infty}(u)e^{iut}du$$

What is the relationship between x(t) and y(t)?

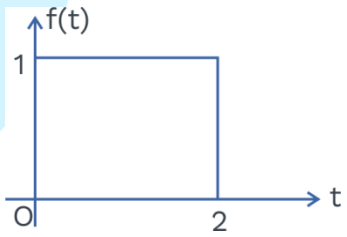
(A) y(t) would always be proportional to x(t)

(B) y(t) would always be proportional to x(t) if x(t) is an even sig.

(C) y(t) would always to be proportional to x(t)

(D) y(t) would proportional to x(t) only if x(t) is sinusoidal function.

Q.28 Find out the Laplace transform and ROC of the given signal?



(A) $\frac{1-e^{-2s}}{s}$, ROC entire plane excluding ∞

(B) $\frac{1-e^{-2s}}{s}$, ROC entire plane excluding $-\infty$

(C) $\frac{1-e^{-2s}}{s}$, ROC entire plane excluding 0

(D) $\frac{1-e^{-2s}}{s}$, ROC entire plane

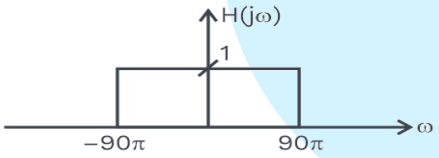
7 Sampling & Filters

Q.1 A signal X(t) = 40 Cos (350πt) is sampled at 700 Hz with sample being taken at time t = –3, What is the value of 6th & 12th sample?

(A) –40, 40 (B) 40, –40

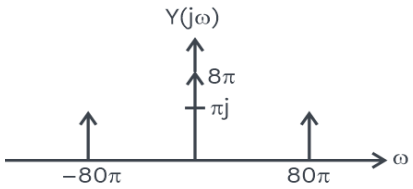
(C) 0, 40 (D) –40, 0

Q.2 A signal X(t) = 4 + Sin (100πt) is samples with sampling interval T_s = 0.0125 and passed through an ideal low-pass filter whose frequency response is shown figure -

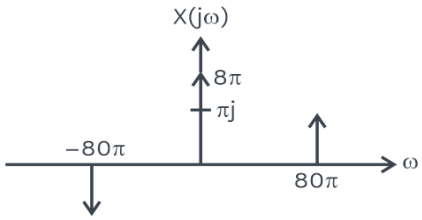


The spectrum of output signal will be

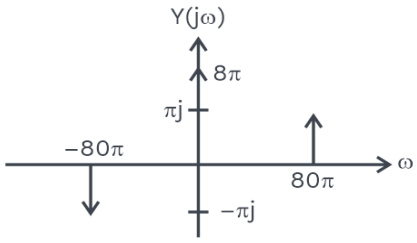
(A)



(B)



(C)



(D) 0

Q.3 A signal with frequency components 75Hz, 125Hz & 175Hz only is sampled at 150 samples/s. The ideally reconstructed signal will have frequency components of-

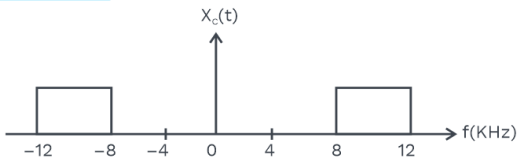
(A) 75Hz only

(B) 75Hz, 25Hz

(C) 5Hz, 275Hz, –25Hz

(D) 50Hz, 75Hz, –25Hz

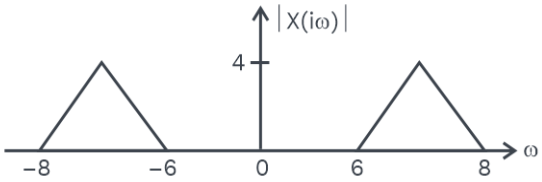
Q.4 Consider a band pass signal $X_C(t)$ is shown below. The minimum sampling rate required for the signal to prevent loss of information –



(A) 4 KHz (B) 4 Hz

(C) 8 KHz (D) 8 Hz

Q.5 What is the minimum Nyquist sampling frequency of the signal X(t) whose amplitude spectrum is shown in the figure below?



(A) 4 rad/sec (B) 8 rad/sec

(C) 6 rad/sec (D) 12 rad/sec

Q.6 Find inverse Laplace transform for t < 0 $\frac{2s-3}{s^2-3s+2}$.

(A) $e^{t}u(t) + e^{2t}u(t)$

(B) $-e^{t}u(-t) - e^{2t}u(-t)$

(C) $e^{t} + e^{2t}$

(D) $-e^{t} + e^{-2t}$

Q.7 Consider a signal $x(t) = (\cos 200\pi t)$ to be sampled. Aliasing take place if the sampling frequency is less than

(A) fs < 400 (B) fs > 400

(C) fs < 200 (D) fs > 200

Q.8 A signal $\frac{1}{t^2}\cos 42\pi t + \cos 10\pi t + t\sin 12\pi t$ is sampled with sampling interval Ts sec. What is the value of Ts. Such that no aliasing will take place?

(A) Ts < 1/21 sec (B) Ts < 21 Sec

(C) Ts < 1/42 sec (D) Ts < 42 sec

Q.9 A signal x(t) having frequency component up to 24 kHz is sampled at a rate of 46 kHz & re constructed through a LPF with unity gain & cut off frequency of 30 Hz which of the following statement is true

(A) The Signal can be reconstructed only for frequencies $|f| < 24$ kHz

(B) The Signal can be reconstructed only for frequencies $|f| < 22$ kHz

(C) The Signal can be reconstructed only for frequencies $|f| > 24$ kz

(D) The Signal can be reconstructed only for frequencies $|f| > 24$ kHz

Q.10 If the bandwidth of a low pass signal g(t) is 5 kHz the bandwidth of g^3 (t) will be -

(A) 5 (B) 125

(C) 15 (D) 25

Q.11 Consider a simple signal processing system shown in figure-



The sampling period of A/D & D/A converters are T_1 = 2.5 m sec 7 T_2 = 5 ms respectively. Input to the system is given $x(t) = 12\sin 200\pi t + 10\cos 250\pi t$ then output y(t) is -

(A) $12\,\mathrm{Sin}(100\pi t) + 10\,\mathrm{Cos}(125\pi t)$

(B) $12\,\mathrm{Sin}(200\pi t) + 10\,\mathrm{Cos}(250\pi t)$

(C) $12\,\mathrm{Sin}\left(\frac{\pi t}{2}\right) + 10\,\mathrm{Cos}\left(\frac{5\pi t}{8}\right)$

(D) $12\,\mathrm{Cos}(100\pi t) + 10\,\mathrm{Sin}(125\pi t)$

Q.12 What is the minimum allowable sampling frequency for following signal?

$X(t) = 50\ \mathrm{Sin}\ (400\pi t)\ \mathrm{Sin}^2\ (1400\pi t)$

(A) $25\ \mathrm{Sin}\ 400\pi t - \frac{25}{2}[\mathrm{Sin}\,3200\pi t - \mathrm{Sin}\,2400\pi t]$

(B) $25\ \mathrm{Sin}\ 400\pi t - \frac{25}{2}[\mathrm{Cos}\,3200\pi t - \mathrm{Sin}\,2400\pi t]$

(C) $25\ \mathrm{Sin}\ 400\pi t - \frac{25}{2}[\mathrm{Cos}\,3200\pi t - \mathrm{Cos}\,2400\pi t]$

(C) $25\ \mathrm{Sin}\ 400\pi t - \frac{25}{2}[\mathrm{Sin}\,3200\pi t + \mathrm{Sin}\,2400\pi t]$

Q.13 What is the ratio of Nyquist Sampling rate of $Sinc^2(250\pi t)$ to the sampling rate of $6Sinc^2(400\pi t)$?

(A) 5/8 (B) 8/5

(C) 3/5 (D) 5/3

Q.14 A signal $x(t) = 3\sin(300\pi t)$ is sampled at a frequency fs = 600 Hz. What is the discreate time signal x(n) generated after sampling? For n = even?

(A) 1 (B) –1

(C) 0 (D) $\frac{\sqrt{3}}{2}$

Q.15 Following signal is sampled at 300 Hz. The fundamental period of the sampled sequence x[n] is

$$x(t) = 40\,Cos\left(\frac{200}{7}\pi t - 125^\circ\right)$$

(A) 21 (B) 1/21

(C) 7 (D) 1/7

Q.16 For input $Sin100\pi t$ an ideal impulse sampling system produces the output

$$y(t) = \sum_{\kappa=-\infty}^{\infty} \frac{1}{8} Sin\left(\kappa \frac{100\pi}{350}\right)\delta\left(t - \frac{k}{350}\right)$$

Where $\delta(t)$ is Dirac delta function. The system is -

(A) nonlinear & time variant

(B) nonlinear & time invariant

(C) linear & time variant

(D) linear & time invariant

Q.17 A band limited signal with a maximum frequency of 10kHz is to be sampled. According to the sampling theorem, the sampling frequency in kHz which is not valid is-

(A) 20 kHz (B) 25 kHz

(C) 30 kHz (D) 15 kHz

Q.18 An ideal square wave signal with time period of 50ms with equal and opposite amplitude, is passed through an ideal low pass filter with cut off frequency 150 Hz. For the output of the given system, which of the given statement is TRUE?

(A) Output is zero

(B) Output contains all odd harmonics of 20Hz

(C) Output contains frequency of 20Hz, 60Hz, 100Hz and 140Hz

(D) Output consists both even and odd harmonics of 20 Hz.

Q.19 For the bilinear transformation method, state with of the statement is TRUE

(A) It is one to many mapping transformations.

(B) Frequency warping does not take place

(C) It has no aliasing effect.

(D) Mapping is linear

Q.20 Find the Nyquist interval for the input signal x(t) = $x(t) = sinc^2(400\,t) * sin^3(300\,t)$

(A) 1/800 sec (B) 800 sec

(C) 900 sec (D) 1/900 sec

Q.21 A continuous-time signal x(t) is bandlimited to 4kHz. The Nyquist rate of $y(t) = x^2(t+4) - x\left(\frac{3}{2}t\right) + x\left(\frac{t}{5}+9\right)$ is _____ kHz.

(Rounded off to the nearest integer)

Q.22 A continuous-time signal x(t) defined as $x(t) = 2\sin(500\pi t - 100)$ is sampled at 2.5 kHz. The fundamental period (N) of the sampled sequence is _____.
(Rounded off to the nearest integer)

Q.23 The impulse response of a filter is given as $h(n) = \underset{\uparrow}{\{1, 2, 1\}}$. The group delay of the

Q.24 A signal $x(t) = 4\cos(240t) + 10\cos(440\pi t)$ is sampled at the sampling frequency of 320 Hz. The sampled signal is passed through an ideal low pass filter with a cut-off frequency of 250 Hz. The total number of frequency components that appear at the output is_____.

Q.25 A signal is given as $x(t) = \text{sinc}(4t) + \cos(3\pi t)$. The signal x(t) is sampled at sampling interval of T_s sec. Aliasing doesn't occur when -

(A) $T_s > 4$ sec (B) $T_s < 8$ sec

(C) $T_s \geq \frac{1}{8}$ sec (D) $T_s \geq \frac{1}{4}$ sec

Q.26 Which of the statement is true for FIR filters?

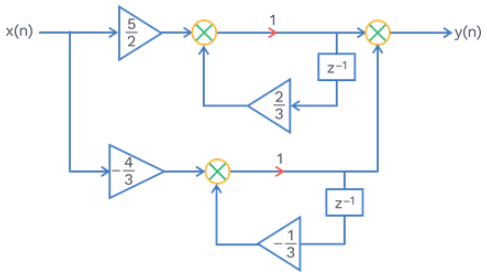
(A) FIR filters are always stable.

(B) FIR filters use feedback from the output.

(C) The number of computations are less in FIR filters.

(D) Phase characteristics are non-linear.

Q.27 The parallel realization of a system using direct form-II is given below. The transfer function of the given system is-



(A) $H(z) = \dfrac{\frac{2}{5}\left(1+\frac{16}{18}z^{-1}\right)}{\left(1-\frac{2}{3}z^{-1}\right)\left(1+\frac{1}{3}z^{-1}\right)}$

(B) $H(z) = \dfrac{\frac{7}{6}\left(1+\frac{31}{21}z^{-1}\right)}{\left(1-\frac{2}{3}z^{-1}\right)\left(1+\frac{1}{3}z^{-1}\right)}$

(C) $H(z) = \dfrac{\frac{2}{3}\left(1+\frac{1}{3}z^{-1}\right)}{\left(1+\frac{2}{3}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)}$

(D) $H(z) = \dfrac{\frac{12}{7}\left(1-\frac{19}{27}z^{-1}\right)}{\left(1+\frac{2}{3}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)}$

Q.28 A continuous time signal x(t) defined as x(t) = 2 sin (500πt – 100) is sampled at 2.5 kHz. The fundamental period (N) of the sampled sequence is _____.

Q.29 A signal x(t) is defined as x(t) = $\text{sinc}^2(300t) + \text{sinc}^3(100t)$

The given signal is sampled at 4 msec.

Which of the following statement(s) is/are TRUE? MSQ

(A) The minimum sampling frequency of x(t) is 600 Hz.

(B) The ideal sampling rate is 8 msec.

(C) The original signal is recovered when the signal is sampled at 4 msec.

(D) The signal undergoes aliasing if the signal is sampled at 4 msec.

Q.30 A continuous time signal x(t) is band-limited to 25 kHz. The Nyquist rate of $x(3t+5)-x\left(\frac{3}{2}t\right)$ is _____ kHz.

Q.31 A signal m(t) = 2 cos 6500 πt + 8 cos 6000 πt + 6 cos 7000 πt is represented by its samples. The minimum sampling rate of the signal considering band pass spectra is_____ kHz. (Rounded off to the nearest integer)

Q.32 The Nyquist rate or Nyquist frequency and the Nyquist interval for $x(t)=\frac{1}{2\pi}\cos(2500\pi t).\cos(1500\pi t)$ is - MSQ

(A) Nyquist frequency = 4000 Hz

(B) Nyquist interval = 200 μsec

(C) Nyquist frequency = 5000 Hz

(D) Nyquist interval = 250 μsec

Q.33 Determine the Nyquist rate for a continuous-time signal

x(t) = 6cos50πt – 20sin300πt + 10cos500πt

(A) 50 (B) 850

(C) 300 (D) 500

8 Z - Transform

Q.1 The ROC (Region of Convergence) of Z-Transform for the discrete time signal $x[n] = 3^n u[n]$ will be

(A) $|z| < 3$ (B) $|z| > 3$

(C) $|z| > 1/3$ (D) $|z| < 1/3$

Q.2 H(z) is a discrete rational transfer function. The condition to ensure both H(z) and it's inverse are stable is

(A) Poles and zeroes needed to be outside the unit circle.

(B) Poles and zeroes needed to be outside the unit circle.

(C) Poles needs to be inside and zeroes needs to be outside the circle.

(D) Poles needs to be outside and zeroes need to be outside the circle.

Q.3 Supposing $x[n] = (\frac{1}{6})^n u[n]$ and the difference equation to be $y(n) - \frac{1}{5}y(n-1) = x[n]$

Assuming initial condition to be at rest, the solution of y(n) for $n \geq 0$ is

(A) $6\left(\frac{1}{5}\right)^n u(n) + 5\left(\frac{1}{6}\right)^n u(n)$

(B) $6(\frac{1}{6})^n u(n) + 6\left(\frac{1}{5}\right)^n u(n)$

(C) $6\left(\frac{1}{5}\right)^n u(n) - 5\left(\frac{1}{6}\right)^n u(n)$

(D) $6\left(\frac{1}{6}\right)^n u(n) - 6\left(\frac{1}{5}\right)^n u(n)$

Q.4 For $x[n] = \left(\frac{1}{6}\right)^{|n|} - \left(\frac{1}{4}\right)^n$, the region of convergence (ROC) for x(n) – Z-transform in z-plane is

(A) $\frac{1}{6} < |z| < 4$ (B) $\frac{1}{6} < |z| < 6$

(C) $\frac{1}{6} < |z| < \frac{1}{4}$ (D) $\frac{1}{4} < |z| < 6$

Q.5 The function of an LTI system is given by

$$H(z) = \frac{1 - \frac{1}{6}z^{-1}}{1 - \frac{1}{3}z^{-1}}$$

The system can have stable inverse, if region of convergence (ROC) of H(z) is defined as

(A) $|z| < \frac{1}{3}$ (B) $|z| > \frac{1}{3}$

(C) $|z| < \frac{1}{6}$ (D) $|z| > \frac{1}{8}$

Q.6 The region of convergence (ROC) of Y(z) for signal $y[n] = \alpha^{|n|}, 0 < |\alpha| < 1$

(A) $|z| < \frac{1}{|\alpha|}$

(B) $|z| > \alpha$

(C) $|\alpha| < |z| < \frac{1}{|\alpha|}$

(D) $|z| < \min\left[|\alpha|, \frac{1}{|\alpha|}\right]$

Q.7 For a discrete time all pass system, the poles are $0.5\angle 0°$ and $4\angle 40°$ The correct statement about the system is

(A) The system is stable only when the impulse response is two sided.

(B) System has constant phase response over all frequencies.

(C) System has constant phase response over the entire z-plane.

(D) System has two more poles at $0.25\angle 40°$

Q.8 X(z) is the z-transform of $x[n]=\delta[n-2]+3\delta[n-6]$

If Y(z) = X(-z) is Z-transform of y(n), then

(A) y(n) = x(-n) (B) y(n) = -x(-n)

(C) y(n) = x(n) (D) y(n) = -x(n)

Q.9 Considering the sequence $x[n] = 2^n u[n] + 3^n u[n]$ where u[n] denotes the unit-step response. The ROC (region of convergence) for the Z-transform of x(n) is

(A) $|z| > 2$ (B) $|z| < 3$

(C) $|z| > 3$ (D) $2 < |z| < 3$

Q.10 Suppose x[n] is an absolutely summable discrete-time signal. It's z-transform is a rational function with two poles and two zeros. The poles are at $z = \pm 2j$. Which one of the following statement is TRUE for the signal x[n]?

(A) It is a finite duration signal.

(B) It is a causal signal.

(C) It is a non-causal signal.

(D) It is a periodic signal

Q.11 If $x[n] = \left(\frac{1}{8}\right)^{|n|} - \left(\frac{1}{5}\right)^{n} u[n]$, then the region of convergence (ROC) of its Z-transform in the z-plane will be:

(A) $\frac{1}{8} < |z|$ (B) $\frac{1}{5} < |z| < 8$

(C) $\frac{1}{8} < |z| < \frac{1}{5}$ (D) $\frac{1}{8} < |z| < 8$

Q.12 Given that

$$x[n] = \begin{matrix} 1 & , & 0 \le n \le 5 \\ 0 & , & \text{elsewhere} \end{matrix}$$

Find the z transform of g[n] if

g[n] = x[n] – x[n – 1]

(A) $1 + z^6$, $|z| > 6$ (B) $1 - z^{-6}$, $|z| > 0$

(C) $1 + z^{+6}$, $|z| > 0$ (D) $1 - z^{-6}$, $|z| > 6$

Q.13 Consider the signal $x(n) = \left(\frac{1}{5}\right)^n u(n-3)$. Evaluate the z-transform of this signal and specify corresponding ROC.

Choose the correct options from the following which is/are corrects? MSQ

(A) $\frac{z^{-3}}{125\left(1-\frac{1}{5}z^{-1}\right)}$ (B) $\frac{z^{-3}}{125\left(1+\frac{1}{5}z^{-1}\right)}$

(C) $|Z| < \frac{1}{5}$ (D) $|Z| > \frac{1}{5}$

Q.14 Using partial fraction expansion and fact that

$$a^n u(n) \xleftrightarrow{z} \frac{1}{1-az^{-1}}, \quad |z| > |a|$$

Find the inverse z – transform of

$$x(z) = \frac{1-\frac{1}{3}z^{-1}}{(1-z^{-1})(1+2z^{-1})}, \quad |z| > 2$$

(A) $\left(\frac{2}{9}\right)u(n) + \left(\frac{7}{9}\right)(2)^n u(n)$

(B) $\left(\frac{2}{9}\right)u(n) + \left(\frac{7}{9}\right)(-2)^{-n} u(n)$

(C) $\left(\frac{2}{9}\right)u(n) + \left(\frac{7}{9}\right)(-2)^n u(n)$

(D) $\left(\frac{2}{9}\right)u(n) - \left(\frac{7}{9}\right)(-2)^n u(n)$

Q.15 Consider the following algebraic expression for the z-transform x(z) of a signal x(n):

$$X(z)=\frac{1+z^{-1}}{1+\frac{1}{3}z^{-1}}$$

Assuming Roc to be $|z| > \frac{1}{3}$, determine the value of x(0), x(1) and x(2).

(A) $x(0) = 0;\ x(1) = -\frac{2}{3};\ x(2) = -\frac{2}{9}$

(B) $x(0) = -1;\ x(1) = \frac{2}{3};\ x(2) = -\frac{2}{9}$

(C) $x(0) = 1;\ x(1) = \frac{2}{3};\ x(2) = -\frac{2}{9}$

(D) $x(0) = -1;\ x(1) = -\frac{2}{3};\ x(2) = -\frac{2}{9}$

Q.16 Let the signal be given as,

$$X(z)=\ln\left(\frac{\alpha}{\alpha-z^{-1}}\right),\quad \text{ROC},\quad |z|>\frac{1}{\alpha}$$

Determine the inverse z transform of the signal.

(A) $\left(-\frac{\alpha^{-n}}{n}\right)\cdot u(n-1)$

(B) $\left(\frac{\alpha^{-n}}{n}\right)\cdot u(n+1)$

(C) $\left(\frac{\alpha^{n}}{n}\right)\cdot u(n-1)$

(D) $\left(\frac{\alpha^{-n}}{n}\right)\cdot u(n-1)$

Q.17 Determine the z-transform of the following signal,

$$x(n)=\left(\frac{1}{2}\right)^n[u(n+4)-4(n-5)]$$

(A) $16z^4 - z^3 + 4z^2 + 2z + 1 + \frac{1}{2z} + \frac{1}{4z^2} + \frac{1}{8z^3} + \frac{1}{16z^4}$

(B) $16z^4 + 8z^3 - z^2 + 2z - 1 + \frac{1}{2z} - \frac{1}{4z^2} + \frac{1}{8z^3} + \frac{1}{16z^4}$

(C) $16z^4 + 8z^3 + 4z^2 + 2z + 1 + \frac{1}{2z} + \frac{1}{4z^2} + \frac{1}{8z^3} + \frac{1}{16z^4}$

(D) $16z^4 + 8z^3 + 4z^2 + 2z + 1 + \frac{1}{2z} - \frac{1}{4z^2} + \frac{1}{8z^3} - \frac{1}{16z^4}$

Q.18 The system has an impulse response $h(n)=\left(\frac{1}{3}\right)^n u(n)$ and the response of the system is $y(n)=\left(\frac{1}{2}\right)u(n)+\left(\frac{1}{4}\right)\left(-\frac{1}{3}\right)^n u(n)$.

Determine the input signal to the given system.

(A) $-\frac{1}{12}\delta(n)+\frac{1}{3}u(-n)+\frac{1}{2}\left(-\frac{1}{3}\right)^n u(-n)$

(B) $-\frac{1}{12}\delta(n)+\frac{1}{3}u(n)+\frac{1}{2}\left(-\frac{1}{3}\right)^n u(n)$

(C) $-\frac{1}{12}\delta(n)-\frac{1}{3}u(-n)-\frac{1}{2}\left(-\frac{1}{3}\right)^n u(-n)$

(D) $-\frac{1}{12}\delta(n)-\frac{1}{3}u(n)-\frac{1}{2}\left(-\frac{1}{3}\right)^n u(n)$

Q.19 Determine the response of the system $y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$ to the input signal $x(n) = \delta(n) - \frac{1}{3}\delta(n-1)$.

(A) $\left(\frac{1}{2}\right)^{-n} u(-n)$ (B) $\left(\frac{1}{2}\right)^{-n} u(n)$

(C) $\left(\frac{1}{2}\right)^{n} u(-n)$ (D) $\left(\frac{1}{2}\right)^{n} u(n)$

Q.20 Assuming zero initial condition the response y(t) of the system given below to a ramp input r(t)



(A) t^2 u(t) (B) $\frac{1}{2}t^2u(t)$

(C) t u (t) (D) e^{-t} u(t)

❖❖❖

Answer Keys

CH-1 Classification and Signals									
1.	B	2.	C	3.	A	4.	A, B	5.	C
6.	D	7.	A, D	8.	0.3678	9.	D	10.	B
11.	C	12.	B	13.	C	14.	B	15.	A
16.	B	17.	D	18.	A	19.	1.666	20.	D
21.	D	22.	A	23.	C	24.	3 J	25.	7

Answer Keys

CH-2 Classification and System									
1.	C	2.	D	3.	D	4.	A	5.	C
6.	A, D	7.	B	8.	C, D	9.	B	10.	A
11.	A, B	12.	B	13.	A, C	14.	C	15.	A
16.	C	17.	B	18.	A, B	19.	A	20.	C
21.	C	22.	C	23.	C	24.	A	25.	A
26.	D	27.	B						

Answer Keys

CH-3 Continuous time Fourier series									
1.	E	2.	C	3.	C	4.	D	5.	D
6.	A	7.	C	8.	C	9.	C	10.	A
11.	A	12.	B	13.	A	14.	C	15.	C
16.	B	17.	B	18.	A	19.	B	20.	B
21.	C	22.	C	23.	B	24.	D	25.	A
26.	B	27.	C						

Answer Keys

CH-4 Discrete time Fourier Series									
1.	D	2.	A	3.	B	4.	A	5.	C
6.	C	7.	A	8.	C	9.	A	10.	A
11.	C								

Answer Keys

CH-5 Fourier Transforms									
1.	C	2.	A	3.	A	4.	A	5.	A
6.	C	7.	A	8.	D	9.	D	10.	B
11.	C	12.	A	13.	A	14.	C	15.	A
16.	C	17.	A	18.	A	19.	C	20.	B
21.	D	22.	A	23.	A	24.	B	25.	C
26.	B	27.	1.33	28.	D	29.	A	30.	A
31.	D	32.	48	33.	C	34.	B	35.	0.25
36.	A	37.	A	38.	B	39.	C	40.	A
41.	2	42.	A	43.	A	44.	B	45.	7
46.	A	47.	B	48.	4	49.	-1	50.	C
51.	B	52.	C						

Answer Keys

CH-6 Laplace transforms									
1.	A	2.	D	3.	C	4.	B	5.	A
6.	D	7.	C	8.	A	9.	A	10.	B
11.	A	12.	D	13.	A	14.	D	15.	C
16.	B	17.	D	18.	B	19.	C	20.	B
21.	C	22.	D	23.	A	24.	D	25.	A
26.	B	27.	B	28.	B				

Answer Keys

CH-7 Sampling & Filters									
1.	A	2.	C	3.	C	4.	C	5.	A
6.	B	7.	C	8.	C	9.	B	10.	C
11.	A	12.	A	13.	A	14.	C	15.	A
16.	C	17.	D	18.	C	19.	C	20.	A
21.	B	22.	10	23.	1	24.	4	25.	D
26.	A	27.	B	28.	10	29.	A, D	30.	150
31.	1	32.	A, D	33.	D				

Answer Keys

CH-8 Z – Transform									
1.	B	2.	A	3.	C	4.	D	5.	B

6.	C	7.	A	8.	D	9.	C	10.	C
11.	B	12.	B	13.	A, D	14.	C	15.	C
16.	D	17.	C	18.	B	19.	D	20.	B